

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

INTEROFFICE COMMUNICATION

TO: Byron Lane, Permits Section, Water Resources Division

FROM: Glen Schmitt, Permits Section, Water Resources Division

DATE: April 11, 2019

SUBJECT: Cadillac Waste Water Treatment Plant (WWTP)
Updated Permit Review Biologist Recommendations
National Pollutant Discharge Elimination System (NPDES) Permit No. MI0023531

Pursuant to your request, we are providing updated recommendations for the NPDES permit reissuance for the Cadillac WWTP. The facility is requesting to continuously discharge a maximum of 3.2 million gallons per day (MGD) of treated municipal wastewater and discharge it through Monitoring Point 001A through Outfall 001 into the Clam River. The Clam River has a 95 percent exceedance, harmonic mean, and 90dQ10 low flows of 2.9, 14, and 3.6 cubic feet per second (cfs), respectively.

Initial recommendations were provided in a memo dated January 1, 2018. The permittee provided comments on the draft NPDES permit and we met with the permittee at Constitution Hall on March 18, 2019 to discuss issues with the draft NPDES permit. Based on the comments provided by the permittee and additional information, we have the following updated recommendations:

Total Copper

- 1) The permittee provided additional final effluent results for total copper in an email dated 3/19/2019. In addition, the permittee indicated the analytical method used for total copper is done in-house using FLAA, SM 3111 B (22nd Edition of Standard Methods). The information included all final effluent data for total copper collected from the Outfall 001 final effluent from January 2013 through January 2019. All the Not Detected data uses a quantification level of 50 ug/l. This quantification level has been determined to be not sufficiently sensitive. According to the permittee in a meeting with MDEQ on 3/18/2019, they conduct the total copper analysis at the Cadillac WWTP. The permittee indicated their equipment can reliably achieve a quantification level of 50 ug/l. MDEQ staff indicate 50 ug/l is no longer sufficient and that future analysis should be conducted using a quantification level between 1 and 10 ug/l. Based on the additional final effluent data for total copper, MDEQ staff did not use the results that reported a quantification level of 50 ug/l as this quantification level is not sufficiently sensitive to determine if total copper is being discharged below Michigan WQS. Therefore, total copper final effluent limitations in the draft NPDES permit should remain unchanged. In addition, the permittee should immediately begin analyzing total copper using an analytical method and quantification level between 1 ug/l and 10 ug/l.

Total Nickel

- 2) The permittee provided additional final effluent results for total nickel in an email dated 3/19/2019. In addition, the permittee indicated the analytical method used for total nickel is done in-house using FLAA, SM 3111 B (22nd Edition of Standard Methods). The information included all final effluent data for total nickel collected from the Outfall 001 final effluent from January 2013 through January 2019. All the Not Detected data uses a quantification level of 50 ug/l. This quantification level has been determined to be sufficiently sensitive. Based on the additional final effluent data for total nickel, there is no longer a reasonable potential for

total nickel to be discharged at concentrations exceeding Michigan WQS. We recommend the draft NPDES permit remove the final effluent limitations for total nickel and replace it with a quarterly monitoring requirement due to elevated levels of total nickel measured in the final effluent. Monitoring for total nickel shall be collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 5.0 ug/l. We do not recommend the draft permit include a monitoring frequency reduction requirement as a sufficient data set for total nickel will be needed during the next reissuance. In addition, the facility already monitors total nickel at a frequency that appears to be greater than quarterly.

Total Selenium

- 3) The permittee provided additional final effluent results for total selenium in an email dated 3/19/2019. In addition, the permittee indicated the analytical method used for total selenium is done in-house using GFAA, SM 3113 B (22nd Edition of Standard Methods). The information included all final effluent data for total selenium collected from the Outfall 001 final effluent from January 2013 through January 2019. All the Not Detected data uses a quantification level of 2.0 ug/l. This quantification level has been determined to be sufficiently sensitive. Based on the additional final effluent data for total selenium, there is a reasonable potential for total selenium to be discharge at concentrations exceeding Michigan WQS. We recommend no changes to the draft NPDES permit for total selenium other than the monitoring frequency be reduced from weekly to monthly. Monitoring for total selenium shall be collected as 24-hour composite samples and analyzed using an approved USEPA analytical method capable of achieving a quantification level of 1.0 ug/l. Monthly monitoring is recommended as the WQBEL is a monthly average limit and the final effluent data evaluated indicated that only one result exceeded the WQBEL.

Bis(2-ethylhexyl)phthalate

- 4) The permittee provided an email dated April 11, 2019 with information from Paragon Laboratories, Inc. that completed the analyses for bis(2-ethylhexyl)phthalate. Specifically, the permittee was going to ask the permittee if the result for bis(2-ethylhexyl)phthalate dated 5/2/2016 was analyzed appropriately and met all lab QA/QC. Information in the April 11, 2019 email indicates the analysis was done appropriately and there were no issues with the QA/QC. The draft NPDES permit includes a monthly average WQBEL for Bis(2-ethylhexyl)phthalate with compliance monitoring weekly. The results evaluated for bis(2-ethylhexyl)phthalate indicate all results are representative results. Based on this information, we recommend no changes to the draft permit with the exception the monitoring frequency can be changed from weekly to monthly. Monthly monitoring should provide a sufficient dataset to evaluate. In addition, the draft permit also includes monitoring frequency reduction language for this parameter. We recommend this parameter to be retained in that condition.
- 5) The recommendations contained herein are based on the Part 4 Water Quality Standards and Part 8 Water Quality-Based Effluent Limit Development for Toxic Substances. Treatment methods or economic concerns have not been addressed. We understand that other considerations may be used in deciding permit limits and requirements.

Please contact me at (517) 284-5599 if you should have any questions regarding the recommendations above.